Applied Algebra

Spring 2026

Instructor: Tuan Tran

Exercise Sheet 10

Deadline: by the end of next Tuesday

Instructions. You are encouraged to work in pairs. When submitting your solutions, list the team members and circle the name of the person responsible for writing up the solutions that week.

Please submit written solutions to **TWO** of the following problems. **Justify your answers.**

Problem 1. The converse of Lagrange's theorem is false: show that A_4, which has order 12, has no subgroup of order 6.

Problem 2. Let G and H be groups. Suppose that for each $h \in H$, we are given an isomorphism $\varphi_h : G \to G$, such that φ_{e_H} is the identity map and $\varphi_{h_1 h_2} = \varphi_{h_1} \circ \varphi_{h_2}$ for all $h_1, h_2 \in H$. Define a product on $G \times H$ by $(g_1, h_1)(g_2, h_2) = (g_1 \varphi_{h_1}(g_2), h_1 h_2)$.

Prove that this makes $G \times H$ into a group. Find its identity and the inverse of (g, h).

Problem 3. Let G be a group and let $g \in G$. Define a function $f : G \to G$ by

$$f(a) = g^{-1} a g \qquad (a \in G).$$

Thus f sends each element of G to its conjugate by g.

Show that f is an isomorphism from G to itself. Show, by example, that f need not be the identity function.

Problem 4. Show that $G \times H$ is Abelian if and only if both G and H are Abelian. Show that the set $\{(g, e) : g \in G\}$ forms a subgroup of $G \times H$.